

Topic:-

Collect information about bimetallic strips and discuss in your class how a fire alarm is made using it.

Introduction:-

Heat is a form of kinetic energy that flows from a body of higher temperature to a body of lower temperature. When solid substances are heated, their atoms vibrate more vigorously and move slightly further apart, causing the substance to expand in length, area, and volume. When cooled, they contract. This universal phenomenon is known as **Thermal Expansion**.

Crucially, different metals do not expand by the same amount even when heated by the exact same rise in temperature. For example, when brass and iron are heated equally, brass expands nearly twice as much as iron! Engineers cleverly take advantage of this difference in expansion rates to invent an ingenious mechanical device called a **Bimetallic Strip**. The prefix 'Bi' means two, and 'Metallic' means made of metals. A bimetallic strip consists of two dissimilar metal strips (typically brass and iron) tightly bonded or riveted together side by side. In this project, we studied how bimetallic strips function as automatic temperature-controlled switches in fire alarms.

❖ Linear Expansion Comparison of Common Metals:-

- **Brass:** High thermal expansion coefficient ($\sim 19 \times 10^{-6} / ^\circ\text{C}$); expands rapidly.
- **Iron / Steel:** Low thermal expansion coefficient ($\sim 12 \times 10^{-6} / ^\circ\text{C}$); expands much less.
- **Result:** Upon heating, brass pulls the strip and forces it to bend into a curve towards the iron side.

Aim & Demonstration Model Setup:-

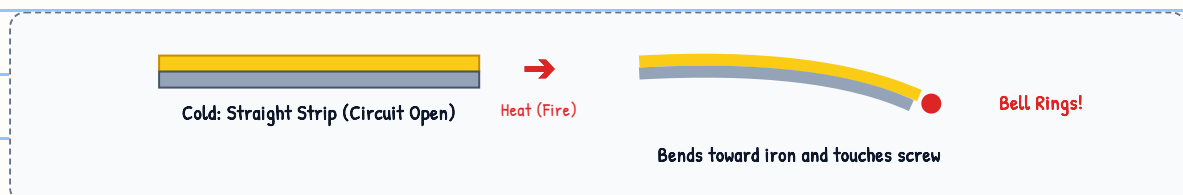
Aim: To collect scientific information about bimetallic strips, understand their thermal bending physics, and build a working model of an automatic fire alarm.

❖ Objectives of the Project:-

- To demonstrate the unequal thermal expansion of brass and iron metals practically.
- To build a complete electric circuit where a bimetallic strip acts as an automatic heat switch.
- To demonstrate how an electric alarm bell rings automatically when fire heat is detected.

❖ Apparatus Required for Working Model:-

- Bimetallic strip (15 cm brass-iron strip riveted together), wooden mounting base.
- Spirit lamp / candle for heat, 6V DC battery, electric buzzer / bell, connecting wires.
- Adjustable brass contact screw on an insulated terminal stand.



Step-by-Step Working Mechanism:-

The automatic fire alarm functions through this elegant mechanical-electrical cycle:

- **1. Normal Room Condition (Circuit Open):** At ordinary room temperature (25°C), the bimetallic strip remains perfectly straight. Its free end sits 2 mm away from the adjustable metal contact screw. The circuit is broken, no current flows, and the alarm bell remains silent.
- **2. Fire Outbreak (Sudden Heat Wave):** If an accidental fire ignites in the room, hot air rises and rapidly heats the bimetallic strip mounted on the ceiling.
- **3. Unequal Expansion Causes Bending:** As temperature climbs past 60°C, the outer brass layer tries to expand much longer than the inner iron layer. Because they are riveted firmly together, brass forces the entire strip to bend and arch downwards towards the iron side.
- **4. Circuit Closes (Alarm Sounds!):** The bending tip touches the contact screw, closing the electric circuit. Current flows from the battery to the electric buzzer, sounding a piercing emergency siren that alerts everyone to evacuate!
- **5. Automatic Reset Upon Cooling:** When the fire is extinguished and the room cools, the brass contracts back to its original length, straightening the strip and turning off the alarm automatically.



Thermal Testing & Expansion Data Table:-

The table below documents our laboratory measurements of bimetallic strip behavior:

<i>Stage / Condition</i>	<i>Temperature Range</i>	<i>Physical Shape of Strip</i>	<i>Electrical Circuit State</i>	<i>Buzzer / Alarm Status</i>
Normal Room Air	25°C	Completely flat & straight	Open circuit (2 mm gap)	Silent (No current)
Mild Heating (Candle 5 cm away)	45°C	Begins curving slightly downwards	Gap narrows to 1 mm	Silent (Still open)
Direct Heat (Simulated Fire)	70°C	Strong curve; tip bends 3.5 mm	Closed circuit (Touches contact)	LOUD SIREN RINGS! (Tring-Tring!)
Flame Removed (Cooling down)	Back to 30°C	Straightens back to flat position	Circuit re-opens automatically	Silences automatically

Scientific Analysis & Discussion:-

Our classroom discussion analyzed the physics that makes bimetallic thermostats so reliable:

- **Why Can't We Use a Single Metal Strip?:** If a switch were made of pure brass alone, heating it would only expand its length by a microscopic fraction of a millimeter (e.g. 0.05 mm). A tiny linear expansion is far too small to bridge an electrical contact reliably. Joining two metals with different expansion coefficients magnifies the tiny linear expansion into a dramatic lateral curve of several millimeters!
- **Curvature Radius Formula:** Physics proves that the radius of curvature of a bimetallic strip is inversely proportional to the temperature rise and the difference in their thermal expansion coefficients: $R \propto 1 / [(\alpha_{brass} - \alpha_{iron}) \times \Delta T]$.
- **Fail-Safe Mechanical Reliability:** Bimetallic fire alarms contain zero microchips, software, or digital sensors that can crash. They operate purely on basic laws of thermodynamics, making them virtually indestructible!

Everyday Applications & Modern Devices:-

Bimetallic strips act as automatic thermal regulators (thermostats) in countless modern appliances:

- **1. Automatic Clothes Iron:** When an electric iron gets too hot, a bimetallic strip bends away from the contact point, cutting power and preventing clothes from burning. As it cools slightly, it snaps back, keeping temperature constant.
- **2. Electric Geysers & Water Heaters:** Automatically cuts heating elements when water temperature reaches 65°C, preventing water boiling and saving power.
- **3. Refrigerators:** Switches the cooling compressor motor on and off to maintain steady cold temperature inside.
- **4. Flashing Indicator Lights:** Car turn-indicator flashers historically used tiny bimetallic strips that heated and clicked on and off rhythmically!

❖ Key Advantages of Bimetallic Thermostats:-

- Completely automatic; requires no human monitoring or complex computer programming.
- Extremely inexpensive to manufacture and lasts for decades without maintenance.

What We Learned & Conclusion:-

- Different metals expand at different rates when subjected to the exact same increase in heat.
- Welding brass and iron creates a bimetallic strip that bends predictably towards the low-expansion metal.
- Bimetallic strips act as simple, reliable, automatic thermal switches in fire alarms and thermostats.
- Practical engineering uses basic physical expansion principles to create lifesaving household safety devices.

Conclusion:-

Our project and classroom demonstration proved how the thermal expansion of metals is brilliantly harnessed to build automatic fire alarms. The working model gave our class a vivid, hands-on demonstration of physics in action: heat was directly converted into mechanical bending, which automatically triggered an electrical emergency siren. Bimetallic strips stand as a timeless testament to human scientific ingenuity.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 14: Measurement and Effects of Heat).
- Standard 8 Science Lab Manual: Thermal Expansion Experiments.
- Understanding Heat & Thermodynamics: Practical Inventions, Balbharti Pune.